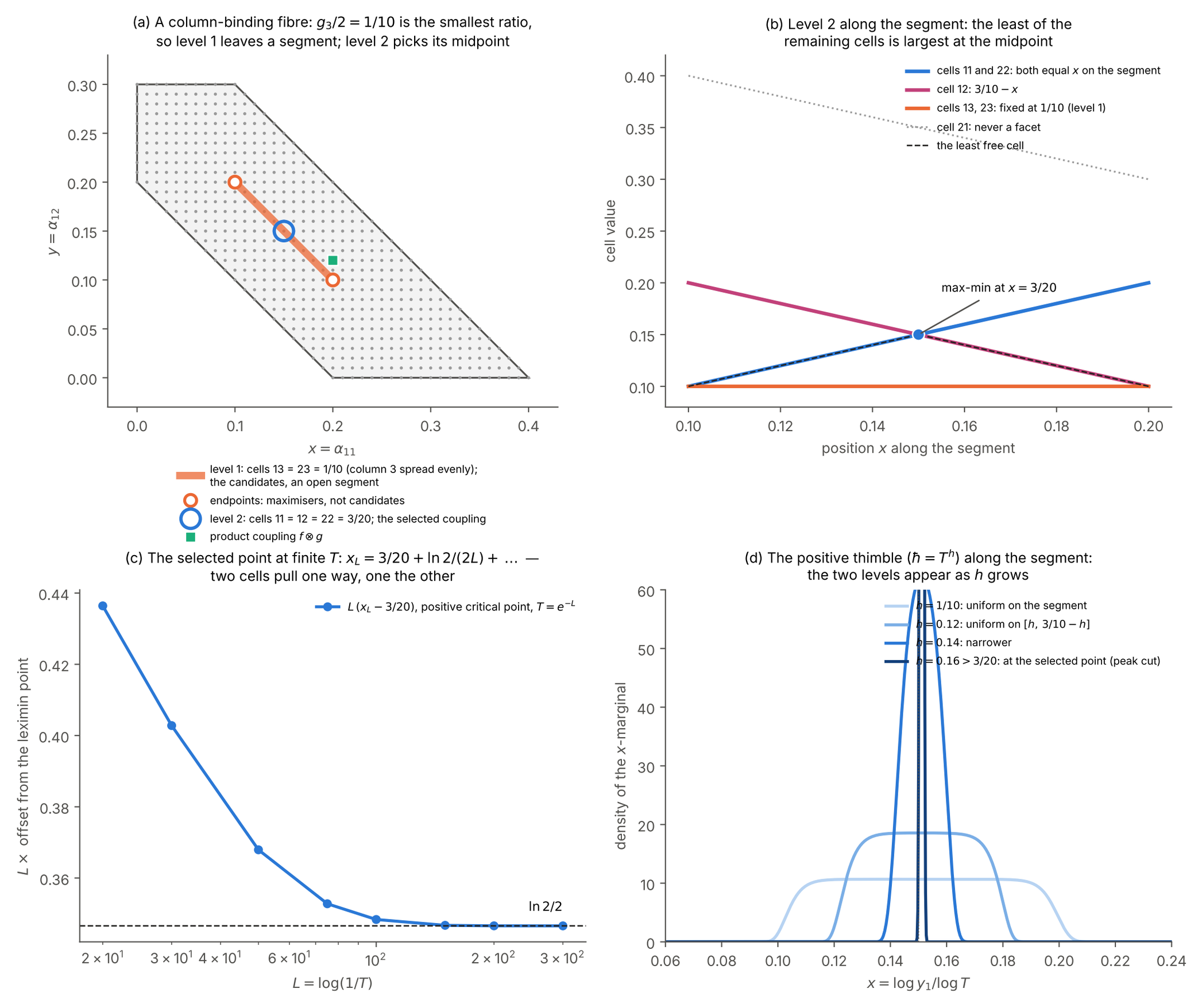
(a) A column-binding fibre: $g_3/2 = 1/10$ is the smallest ratio, so level 1 leaves a segment; level 2 picks its midpoint
$y = \alpha_{12}$
$x = \alpha_{11}$
level 1: cells 13 = 23 = 1/10 (column 3 spread evenly); the candidates, an open segment
endpoints: maximisers, not candidates
level 2: cells 11 = 12 = 22 = 3/20; the selected coupling
product coupling $f \otimes g$
(b) Level 2 along the segment: the least of the remaining cells is largest at the midpoint
cells 11 and 22: both equal x on the segment
cell 12: $3/10 - x$
cells 13, 23: fixed at 1/10 (level 1)
cell 21: never a facet
the least free cell
max-min at $x = 3/20$
cell value
position x along the segment
(c) The selected point at finite T: $x_L = 3/20 + \ln 2/(2L) + \ldots$ — two cells pull one way, one the other
L (x_L − 3/20), positive critical point, $T = e^{-L}$
$L \times$ offset from the leximin point
ln 2/2
$L = \log(1/T)$
(d) The positive thimble ($\hbar = T^h$) along the segment: the two levels appear as h grows
$h = 1/10$: uniform on the segment
$h = 0.12$: uniform on $[h, 3/10 - h]$
$h = 0.14$: narrower
$h = 0.16 > 3/20$: at the selected point (peak cut)
density of the x-marginal
$x = \log y_1 / \log T$